

Grace — the CKM/PMNS cross-shelf overlap-matrix skeleton (task #76, 2026-08-05, K1181)

Mixing = $\langle \text{up} | \text{down} \rangle$ overlap of which SHELVES the modes sit on (NOT their masses). The up-mass FK-failure (Elie 5060) kills “up-masses are FK-forced,” NOT “mixing is computable.” I build the skeleton + selection rule; Elie fires the FK-overlap values BLIND (I do not compute/tune them).

★ UPDATE (K1183, five-seat convergence) — the mixing is W-MEDIATED, not a naive overlap

Up on EVEN shelves $\{0,2,4\}$, down on ODD shelves $\{1,3,5\}$ = opposite parity under the boundary fold (mirror $\det=-1$). So the naive inner product $\langle \text{up} | \text{down} \rangle = \langle \text{even} | \text{odd} \rangle = 0$ by parity — even and odd harmonics are orthogonal. The physical mixing therefore CANNOT be a plain overlap; it must go through the operator that carries the parity flip: $\text{CKM} = \langle \text{up } \{0,2,4\} | J_W | \text{down } \{1,3,5\} \rangle$ — the weak charged current J_W bridges the parity gap. The unification (three names, one object): the parity-bridging operator = the weak charged current J_W = the $\text{up} \leftrightarrow \text{down}$ isospin-raising operator. And this is exactly the SM definition of CKM (the charged-current W-coupling) — BST *forces* the CKM to be a W-current matrix element because the direct overlap vanishes. This is WHY CKM is small and near-diagonal: mixing is a *current matrix element* (one J_W insertion for the diagonal, higher J_W powers for the off-diagonal $\rightarrow$ Wolfenstein suppression), not a free overlap. The vanishing overlap is the feature, not a bug. Five independent seats (Lyra charge-conservation, Grace selection-rule, Elie S^1 -charges, Cal §287 degree-orthogonality, Keeper parity-flag) converged on this same turn.

G1 — the skeleton (now read with J_W between the shelves)

Up-type on EVEN degrees $u=0, c=2, t=4$; down-type on ODD degrees $d=1, s=3, b=5$. $\text{CKM}_{ij} = \langle \text{up}_i | \text{down}_j \rangle$, a cross-parity FK overlap. Degree gap $\Delta_{ij} = 2(j-i)+1$ (always ODD = the parity selection rule):

	d (1)	s (3)	b (5)
u (0)	$ \Delta =1$	$ \Delta =3$	$ \Delta =5$
c (2)	$ \Delta =1$	$ \Delta =1$	$ \Delta =3$
t (4)	$ \Delta =3$	$ \Delta =1$	$ \Delta =1$

- Selection rule: overlap nonzero only for Δ odd (even $\leftrightarrow$ odd) — automatic from the parity grid. FK form $N(w)^{\{n_C/2\}}$ supplies the magnitude.
- Structural check (real): the diagonal is $|\Delta|=1$ (minimal cross-parity gap) $\rightarrow V_{ud}, V_{cs}, V_{tb} \approx 1$ (the largest elements, as observed).
- Honest flag (NO retrofit): the raw $|\Delta_{deg}|$ is ASYMMETRIC ($V_{us} |\Delta|=3$ vs $V_{cd} |\Delta|=1$) while observed CKM is near-symmetric $\rightarrow$ the FK overlap INTEGRAL must supply the symmetrization. That is exactly what Elie fires blind; I do not tune it.
- PMNS analog: $\langle \text{charged-lepton (odd } \{1,3,5\}) | \text{neutrino (even } \{0,2,4\}) \rangle$ — SAME cross-parity structure, but the neutrino side STARTS AT $k=0$ (the massless mode). That extra zero-degree partner is a plausible structural reason PMNS mixing is LARGE where CKM is small — a lead for Elie’s fire, not banked.

G2 (decisive) — saturation moves the NORM, not the address (LEANS computable)

Does top-Yukawa saturation move the mass-norm or distort the wavefunction? Leans NORM: - Corpus (OP-4/K765/K766): “a Yukawa IS a Born overlap; $|y| \leq 1$ is KINEMATIC” — saturation caps the MASS-overlap MAGNITUDE (the norm), a bound on the overlap value, not a re-addressing of the mode. - The up ADDRESS $\{0,2,4\}$ is forced by charge-parity (crux 1, $m=3|Q|$, T2470) — which saturation does NOT touch. - $\square$ saturation re-sets the up MASS-mechanism (FK-norm $\rightarrow$ boundary-overlap) while the ADDRESS stays $\{0,2,4\} \rightarrow$ the mixing $\langle \{0,2,4\} | \{1,3,5\} \rangle$ is computable from forced addresses. - CAVEAT (flag, do NOT bank): needs Lyra/Elie to confirm the up-wavefunction is not pulled off $\{0,2,4\}$ by Shilov boundary-localization. If it stays, the off-diagonal fire is clean. - UPDATE (K1183) — now passing on TWO independent routes: (1) my corpus trace (saturation caps the

norm, not the address); (2) Lyra's — the Higgs is electrically NEUTRAL, so it can reset the top's mass-norm but cannot move it to a different-charge address (address = charge = m, fixed). Two routes agree: address preserved. Only residual: does making the top heavy reshape its RADIAL profile enough to matter (Lyra/Elie rule). G2 is near-closed. Cal sub-flag: with up-mass no longer pinning the up-tower ORDERING, the ordering is forced by charge-parity and CONFIRMED BY THE MIXING itself (as the mass confirmed the down-tower).

G3 — neutrino $k=0$ shelf reconciled with the forced massless

Neutrino even grid starts at $k=0$. The $k=0$ FK mode = $(\nu)_0 = 1$ (empty Pochhammer) = null/ground weight = massless = the forced $m_1=0$ (pred_003). Reconciles with my F93 ν -varying $\{5/2, 3/2, 0\}$: degree- $k=0$ and Wallach- $\nu=0$ BOTH name the zero mode; the massless lightest is robust across both readings. Whether the degree-grid and ρ -grid are one structure re-read is open (Lyra/Elie) — flagged, not banked.

Ledger state

The 7 mixing parameters STAY Identified. This skeleton makes them COMPUTABLE (the linear-algebra structure), but they flip to Derived only when: G2 confirmed (up stays on $\{0, 2, 4\}$) + Elie fires the off-diagonal FK overlap blind + Keeper counts independent-N + it reproduces without tuning. Forced-address-or-nothing.

★ THE WEAK-CURRENT MATRIX from cohomology (K1184, task) — pure linear algebra on D_{IV}^5

Q^5 cohomology ring $\mathbb{Z}[h]/h^6$ (T1929, degree-graded). Up = EVEN classes $\{h^0, h^2, h^4\}$; down = ODD $\{h^1, h^3, h^5\}$. J_W = the degree- ± 1 operator (cup-product-with- h / adjoint) — the UNIQUE map connecting even $\leftrightarrow$ odd classes. The weak charged current IS this degree-1 cohomology map. $CKM_{ij} = \langle up_i | J_W^{\Delta deg} | down_j \rangle$. Minimal-power matrix:

	d(1)	s(3)	b(5)
u(0)	J_W^1	J_W^3	J_W^5
c(2)	J_W^1	J_W^1	J_W^3
t(4)	J_W^3	J_W^1	J_W^1

ANGULAR SKELETON = STRUCTURAL-DERIVED (zero up-mass input) — but only the CRUDE shape (honest split): - Derived (angular): (1) diagonal = $J_W^1 = O(1) \rightarrow$ why CKM diagonal ≈ 1 ; (2) mixing exists ONLY through $J_W \rightarrow$ why it's small AT ALL (direct overlap vanishes by parity, even $\nleftrightarrow$ odd); (3) corners most-suppressed ($1-3 = J_W^5$). These three are forced by parity-grading alone. - NOT from the angular skeleton (rides the radial $\rightarrow$ Identified): the raw J_W -powers are too crude for the detailed pattern — they give $V_{us}=J_W^3$ and $V_{cb}=J_W^3$ (same power) though observed $V_{us} \gg V_{cb}$ (λ vs λ^2), and $V_{us}=J_W^3$ vs $V_{cd}=J_W^1$ (asymmetric) though the 1-2 block is symmetric to 0.1%. So the Wolfenstein λ -grading, the 1-2/2-3 near-symmetry, and the exact values ALL ride the radial FK overlaps — Elie's blind fire. The angular skeleton gives the *shape* (near-diagonal, parity-small); the radial gives the *numbers*. I do NOT over-claim the hierarchy from the powers. - 1-3 corner = the discriminator (Cal §288): $V_{ub}=J_W^5$ vs $V_{td}=J_W^3$, different powers $\rightarrow$ intrinsically asymmetric; a symmetric-by-construction ansatz FAILS there. Observed: 1-3 asymmetric by $\sim$ factor 2, 1-2 sym 0.1%, 2-3 $\sim$ 2%. The fire must reproduce that from the radial tail. - Honest cost — RETRACTED (K1186, Casey's insight verified): I charged the up-tower ORDERING as one extra input. It is FREE. T2515 (corpus, Tier I; "Grace unification"): the up-tower is $y=\exp(-\text{geodesic distance to boundary})$, one shell = $\ln(N_{\max}) \rightarrow \alpha$ per shell — top AT the boundary (saturated $y_t=1$), charm one shell in ($y_c=\alpha$), up two shells in. So higher shelf = closer to boundary = heavier is a corpus-forced monotonicity $\rightarrow$ the up-ordering ($u=0, c=2, t=4$ by mass) is FORCED by boundary-distance, the SAME mechanism that makes the top anomalously heavy. The kludge and the ordering are one mechanism. So the down-tower is forced by FK-norm, the up-tower by the boundary ladder — BOTH forced, NO extra input. On the fire, the 7 params can go FULLY Derived (no residual input cost). - REFINEMENT (K1189, Cal §291 — my " α per shell" was too clean): the up-ORDERING stays free (monotonicity: closer to boundary = heavier), but the

per-rung FACTOR is NOT a uniform α -per-shell — charm $\rightarrow$ top $\approx \alpha$ ($t/c=136\approx\alpha^{-1}$ [?]) but up $\rightarrow$ charm $\approx 4.2\times\alpha^{-1}$ ($c/u=577$, anomalously steeper). So each RUNG VALUE must be independently shown forced, not assumed- α . Ordering forced; per-rung values are the open piece the dial-free fire must show forced (not fitted).

Tier split (K1184): angular skeleton (shape) = Structural-Derived, no up-mass dependence — bankable as the *shape* of the mixing. The seven VALUES ride the radial $\rightarrow$ Identified until Elie's blind fire reproduces them (octant chosen first, 1-3 corner is the make-or-break).

★ THE ORDERED PRODUCT (K1187, task #77, Casey's non-commutativity insight) — the possible route to the VALUES

EXACT linear-algebra fact: mixing IS the non-commutativity — if $[M_{\text{mass}}, J_W]=0$ (shared basis), CKM=identity, nothing mixes. Commit $\rightarrow$ emit ordering (the item-10 record cycle): mass = commit (stored diagonal), J_W = emit (transition) $\rightarrow$ CKM = $U_{\text{up}}^\dagger \cdot J_W \cdot U_{\text{down}}$ (mass-basis first, weak transition second). The order is the substrate cycle, not a choice — the flavor sector IS the commit cycle running in fermion space.

The ordered product from FORCED operators: down ladder (FK-norm {1,3,5}: 1:20:840), up ladder (boundary α -per-shell {0,2,4}: $\alpha^2:\alpha:1$, T2515), J_W (one-degree step). All three forced.

LEADING-TERM CHECK (real, but CALIBRATED K1189): Cabibbo $\lambda = \sqrt{(m_d/m_s)} = 1/\sqrt{20} = 0.2236$ vs obs 0.2243 ($<1\sigma$) — passed Cal's pre-registered razor (§291: “must give $1/\sqrt{20}$, not just near 0.22”), four seats. BUT this λ IS the Gatto relation (Gatto 1968, $\sqrt{(m_d/m_s)}$), ALREADY BANKED in the corpus. [?] it VALIDATES the ordered-product mechanism but banks NO new number. What is genuinely BST: (a) forces the input ratio $m_s/m_d=20$ from geometry (not measured); (b) DERIVES the geometric-mean texture from the degree-1 cohomology operator = EXPLAINS why Gatto holds; (c) grounds the order in the commit cycle. CLAIM EXACTLY “BST forces + explains Gatto”, NOT “BST derives the Cabibbo” (referee would pounce). Nothing new banked; confidence in the route went up.

HONEST TIER: SOLID/EXACT — mixing=non-commutativity + commit $\rightarrow$ emit order. SOLID — leading $\lambda=1/\sqrt{20}$ forced. HYPOTHESIS (not a theorem): that the full ordered product forces λ^2 , λ^3 , and the factor-2 corner from (α -per-shell up-ladder) $\times$ (one-degree J_W) with no tuning. Ingredients all forced (down FK, up α T2515, J_W); the open question is whether their PRODUCT reproduces the tower or needs a hidden assembly coefficient (Cal guards). Elie's dial-free fire decides. If forced $\rightarrow$ values come from the SEQUENCE (strongest possible outcome, 7 params $\rightarrow$ Derived). If a coefficient is tuned $\rightarrow$ shape stays Structure-Derived, values Identified. I build the structure; I do NOT claim the tower until the product is computed.

★ G4 — THE NEUTRINO RECONCILIATION (task #76 open part, 2026-08-12): small-CKM/large-PMNS split is FORCED by the SOURCE structure

The two mixings share the SAME cross-parity skeleton (G1), so the small-vs-large split cannot come from the skeleton — it comes from the mass-diagonalizing rotations, which come from the SOURCES: - Linear-algebra fact: CKM = $U_{\text{up}}^\dagger \cdot U_{\text{down}}$, PMNS = $U_{\text{e}}^\dagger \cdot U_\nu$. Same source $\rightarrow$ aligned U's $\rightarrow$ they cancel $\rightarrow$ mixing ≈ 1 (small). Different sources $\rightarrow$ uncorrelated U's $\rightarrow$ mixing = $O(1)$ angles (large). - QUARKS — small CKM, FORCED: up & down are BOTH Dirac overlaps on the SHARED rank-1 condensate O (F603, flagship — O = SO(5) vector, the one condensate direction). So $U_{\text{up}} \approx U_{\text{down}} \approx U_O \rightarrow$ CKM $\approx 1 + (\delta_d - \delta_u) = 1 + \text{small}$. The shared O forces near-diagonal CKM. - LEPTONS — large PMNS, FORCED: charged leptons are Dirac from O (like quarks), but the neutrino mass is MAJORANA from a DIFFERENT source φ (T1985: no $\nu_R \rightarrow$ the Dirac operator $\bar{\nu}_L \nu_R$ cannot exist $\rightarrow$ mass comes from a distinct Majorana operator, $\varphi \neq O$). U_ν diagonalizes φ , uncorrelated with $U_O \rightarrow$ PMNS = $U_O^\dagger \cdot U_\varphi$ = mismatch of two unrelated bases $\rightarrow$ LARGE. - Numerical illustration (mechanism, not the angles): shared-O quark toy $\rightarrow$ CKM off-diag ~ 0.23 (small); different- φ neutrino toy $\rightarrow$ PMNS off-diag ~ 0.59 (large). Swapping ν -source to O collapses PMNS to small — confirming the LARGE is caused by $O \neq \varphi$, not by tuning. - Corroboration (target-innocent): the ν even-grid starts at $k=0$ (massless $m_1=0$, T1985/pred_003) with NO ν_R partner — the extra unpaired zero-mode is the same “different structure” fact, a second forced reason U_ν is unconstrained by U_e .

TIER: the small-CKM/large-PMNS SPLIT is STRUCTURAL-DERIVED — forced by (shared source O for quarks, F603) + (different Majorana source φ for ν , T1985 no- ν_R). This is the “one Peirce statement” (Cal §137) made concrete: same skeleton, opposite mixing size, because the sources differ. The exact angles ride the radial FK/Majorana overlaps — Elie’s blind fire, NOT claimed here. #76 neutrino-reconciliation: CLOSED at the structural (split-forced) tier.